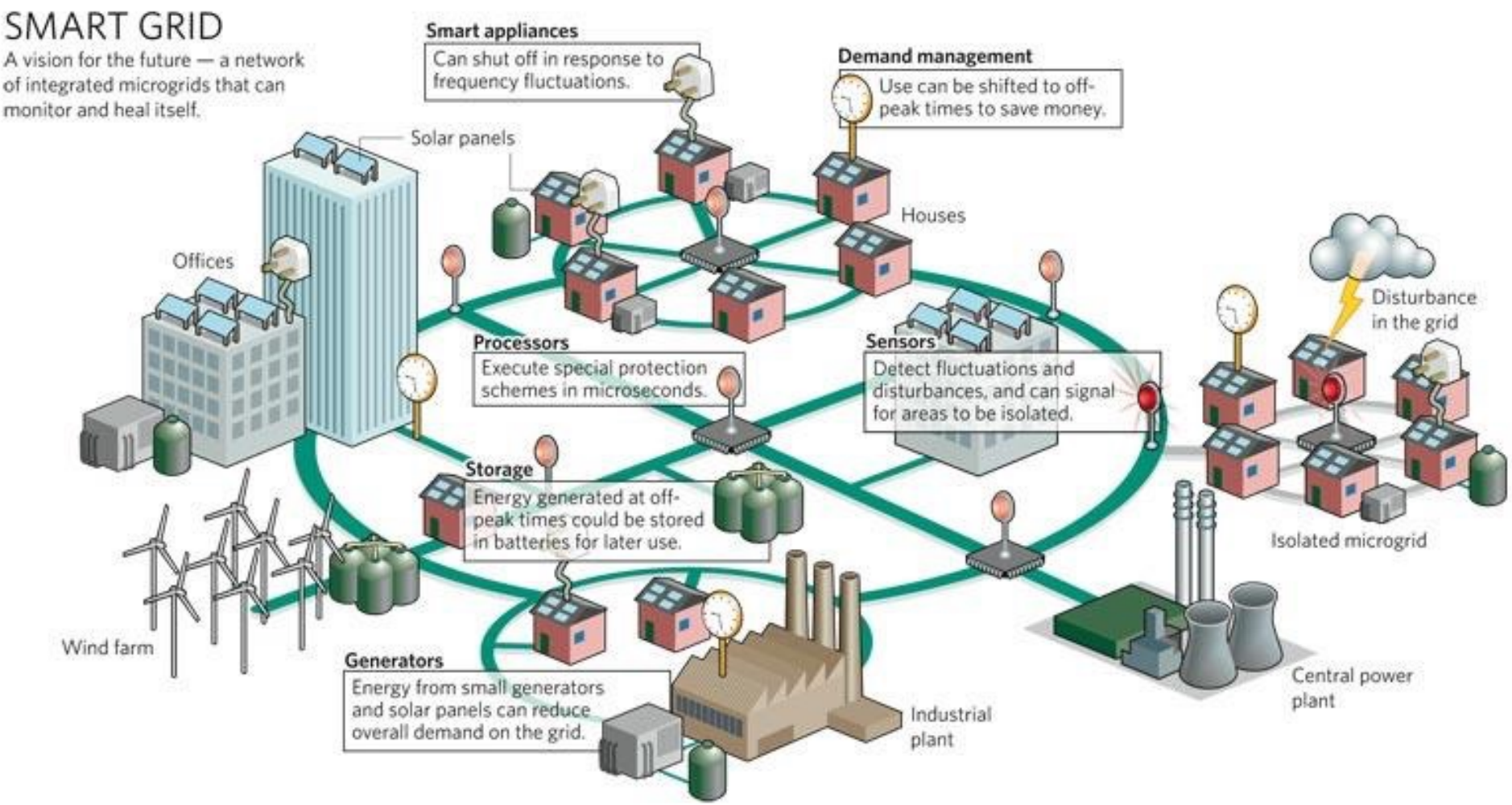
Today's PBL

UK domestic energy use

Solar Energy

Wind (electricity transmission)

Smart Grids



- A key concept is *demand management* in addition to *supply management*.

PBL 4

- What percentage of UK Carbon footprint is from domestic energy usage?
- What can UK homeowners do to reduce emissions?

See Solar Lecture

Solar Power

PHAS0050 : Climate & Energy

Prof. Cham Ghag & Prof. Chris Howard

1. Introduction
2. Solar Power Potential Yield
3. Solar Hot Water Heating
4. Photovoltaics
5. Solar Cell Efficiencies
6. Economics of Solar Cells
7. Solar Concentrators
8. Summary

Problem Based Learning Session 4

PHAS0050 : Climate & Energy

2019 Q7

(b) The bandgap in germanium is 0.67 eV. What is the longest photon wavelength that can induce a current in a germanium PV cell? **[2]**

Consider photons with energies above and below 0.67 eV. What fraction of the incident photon's energy can be captured in a germanium PV cell in each case? **[4]**

[Planck's constant $h = 6.6 \times 10^{-34}$ Js; the speed of light $c = 3.0 \times 10^{8}$ m/s; 1 eV = 1.6×10^{-19} J.]

See Solar Lecture

Problem Based Learning Session 4

2019 Q7

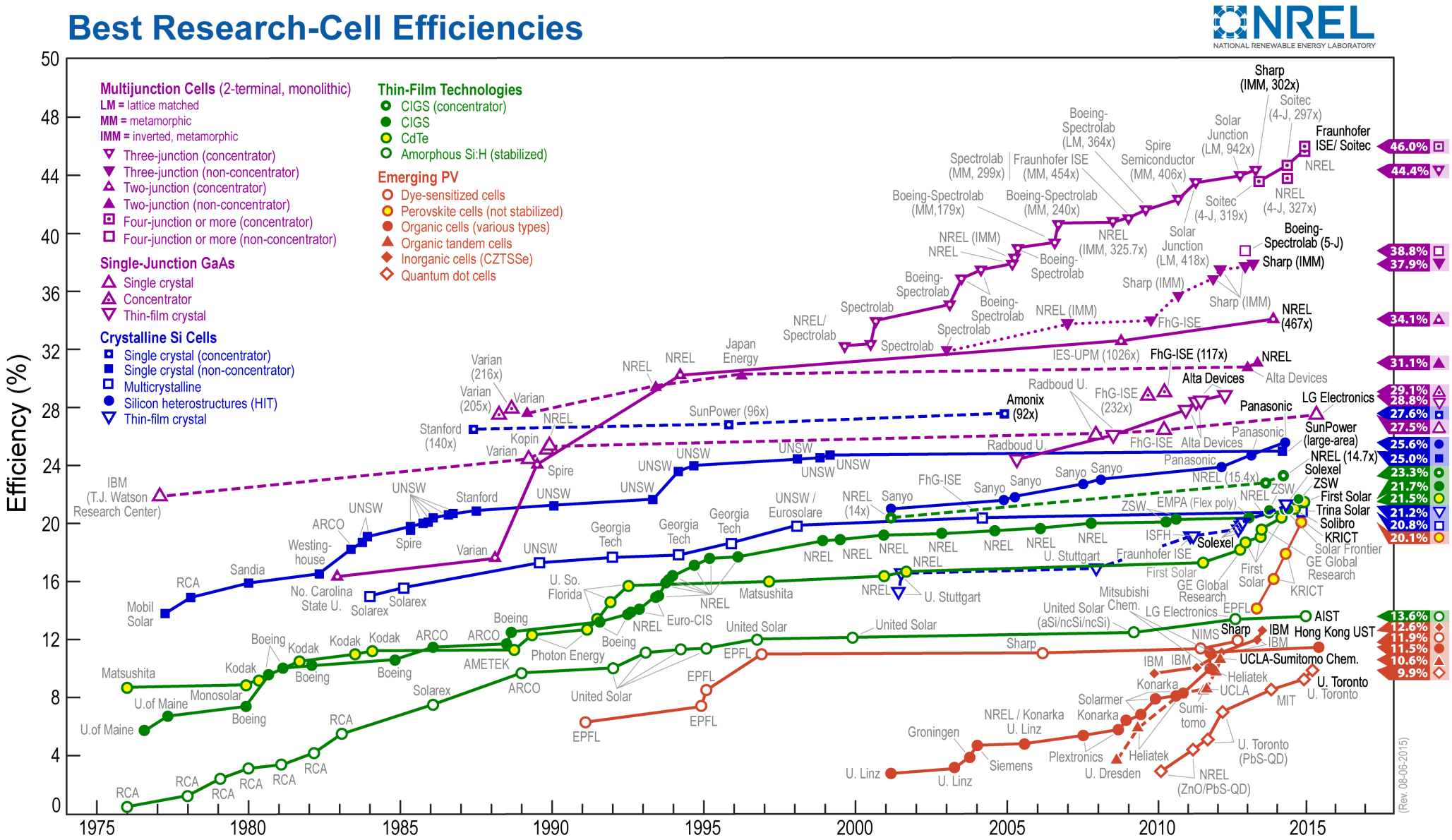
(e) Suppose that solar radiation is optically concentrated (using mirrors or lenses) by a factor of 100 before falling onto a silicon PV cell which has an open-circuit voltage under normal (non-concentrated) illumination of 0.66 V. Assuming the same current and "fill-factor", estimate the increase in efficiency of the PV cell under concentrated sunlight. You can assume that $T = 300$K. **[5]**

Numerical data:

Boltzmann's constant $k = 1.381 \times 10^{-23}$ J/K; electron charge $e = 1.602 \times 10^{-19}$ C.

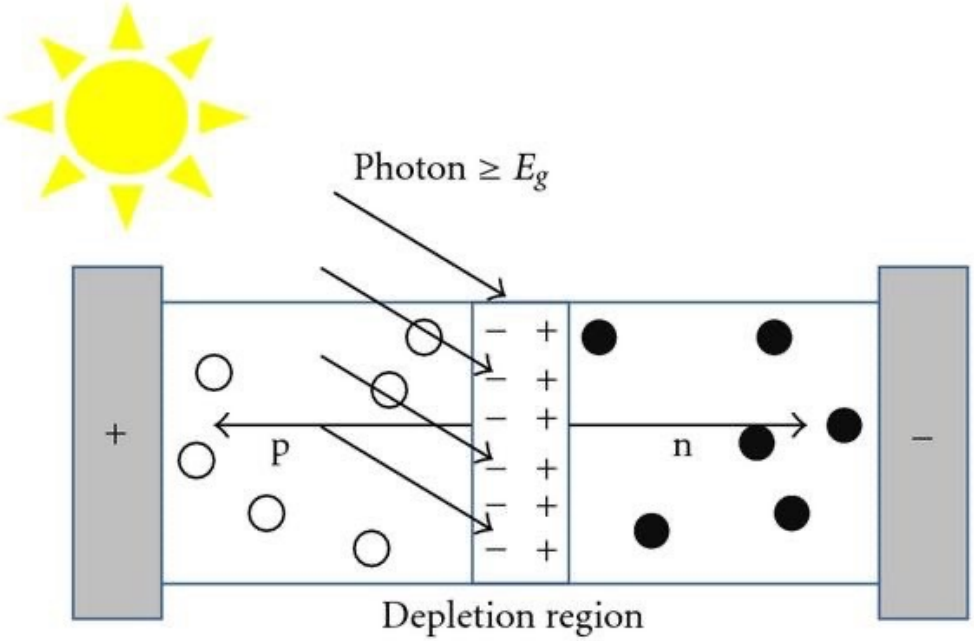
See Solar Lecture

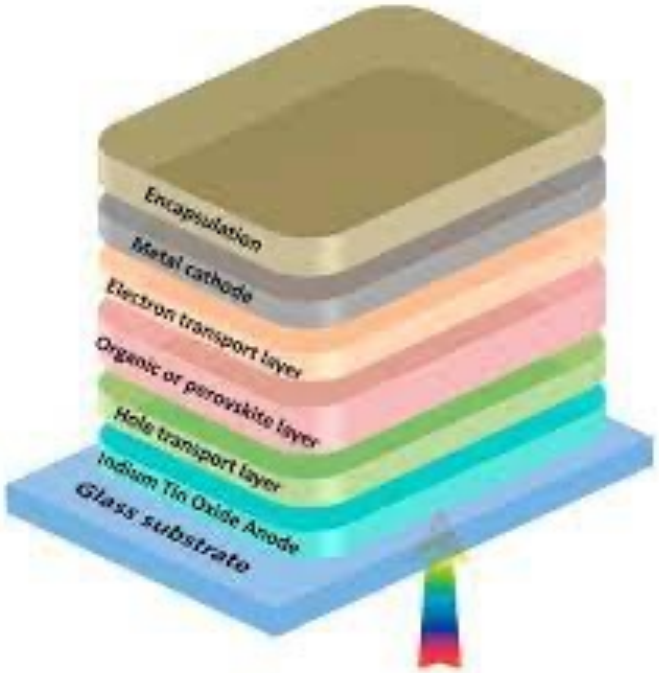
Efficiency of a Solar Cell



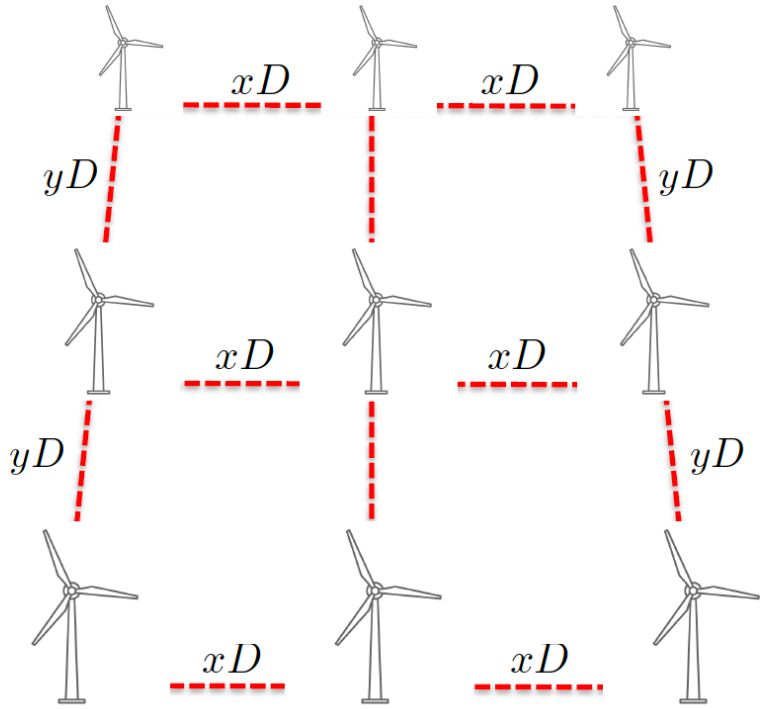
See Solar Lecture

Photovoltaics





Wind Farm Capacity



Typical separations :

$$x = 4; \; y = 7$$

Wind farm power area density :

- 5 MW rated turbines
- Capacity factor 0.35
- 115m diameter turbines

$xD = 4 \times 115\text{ m} = 0.46\text{ km}$

$yD = 7 \times 115\text{ m} = 0.805\text{ km}$

$\Rightarrow$ Area occupied by each turbine $= 0.37\text{ km}^2$

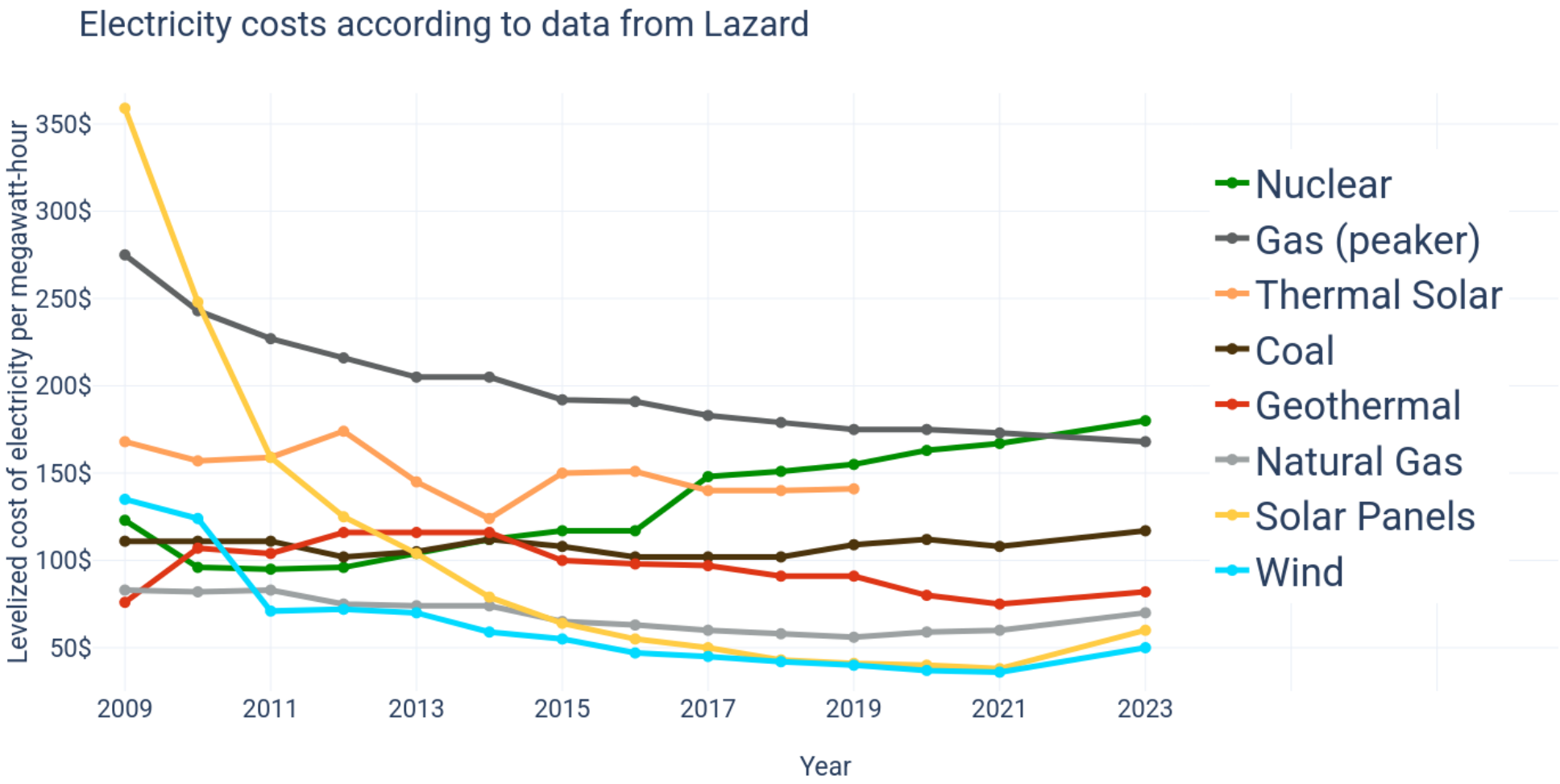
$\Rightarrow$ Power area density $=$

$$\frac{1}{0.37} \times 5 \times 0.35 = 5\text{ MW/km}^2$$

Photovoltaic Power Densities

Q1. Calculate the power per km^2 for a solar farm that has an incident solar power density of 2000 $kWh/m^2/yr$. Assume panels have a 20% efficiency

Wiki - Levelized_cost_of_electricity



https://en.wikipedia.org/wiki/Levelized_cost_of_electricity#/media/File:Electricity_costs_in_dollars_according_to_data_from_Lazard.png

Problem Based Learning Session 3

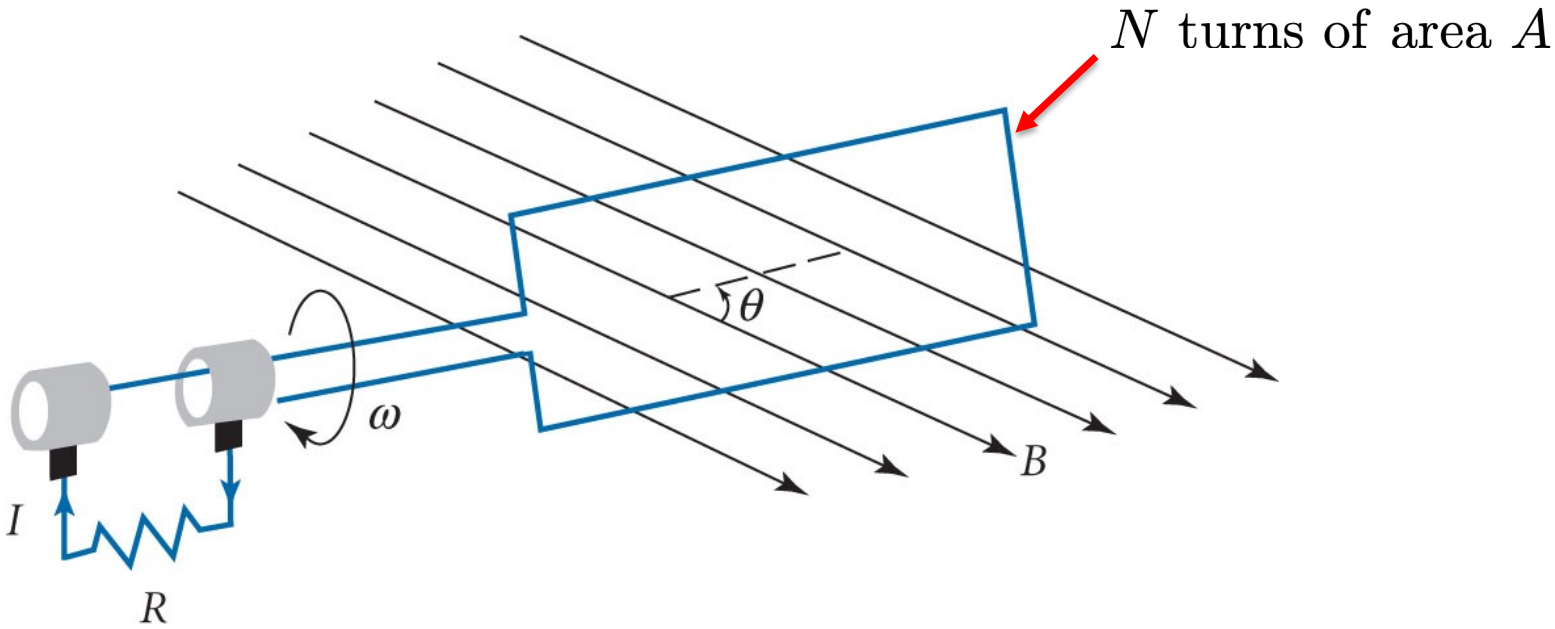
2018 Q10

(c) Describe how power from a remote wind or tidal turbine array may be delivered to the national UK electricity grid. Include a description of how low frequency turbine rotors can deliver much higher frequency alternating current (AC) electricity, and why it may be advantageous to transmit power over High Voltage Direct Current (HVDC) transmission lines. **[4]**

An HVDC line has a resistance of 0.01 Ohm/km. Calculate the fractional ohmic energy loss if this HVDC line is used to transmit 2500 MW of electrical power at a voltage of 500 kV over a distance of 600 km. **[4]**

What will the fractional ohmic energy loss of the same HVDC cable be, if it delivers the same power but at the higher voltage of 800 kV? **[2]**

Generating Electricity

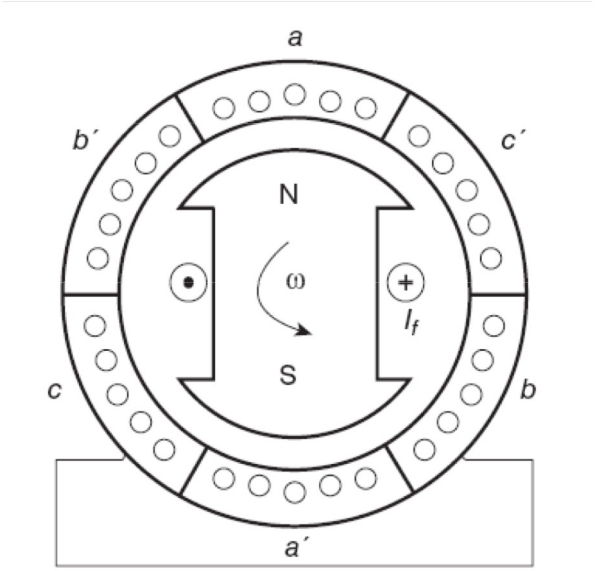


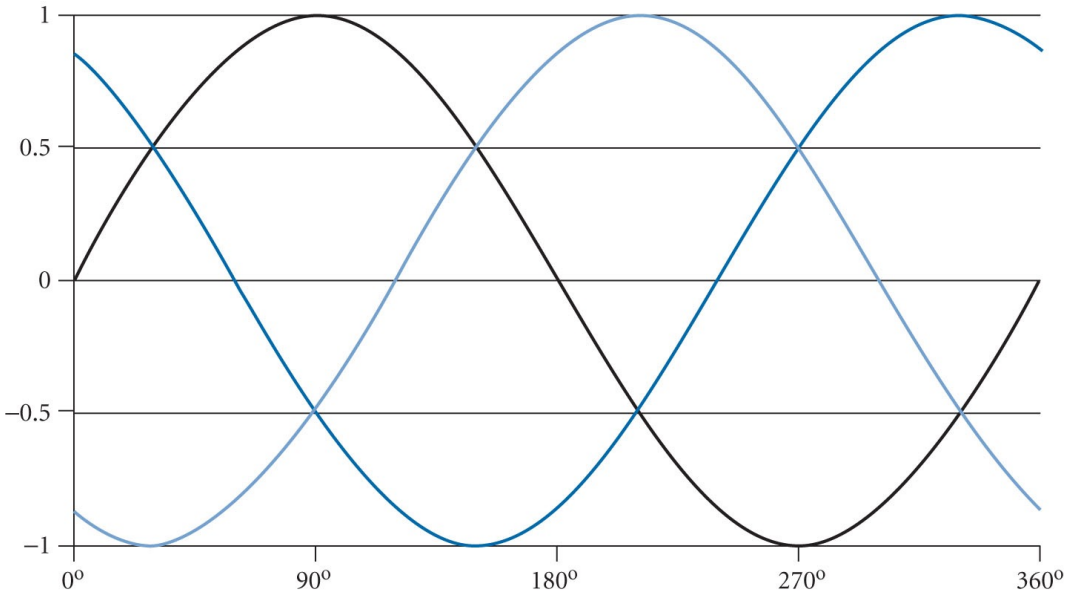
$$\text{Magnetic flux } \phi = NBA\cos\theta = NBA\cos\omega t$$

$$\text{Faraday}: V = -\frac{d\phi}{dt} = NBA\omega\sin\omega t$$

Generating Electricity

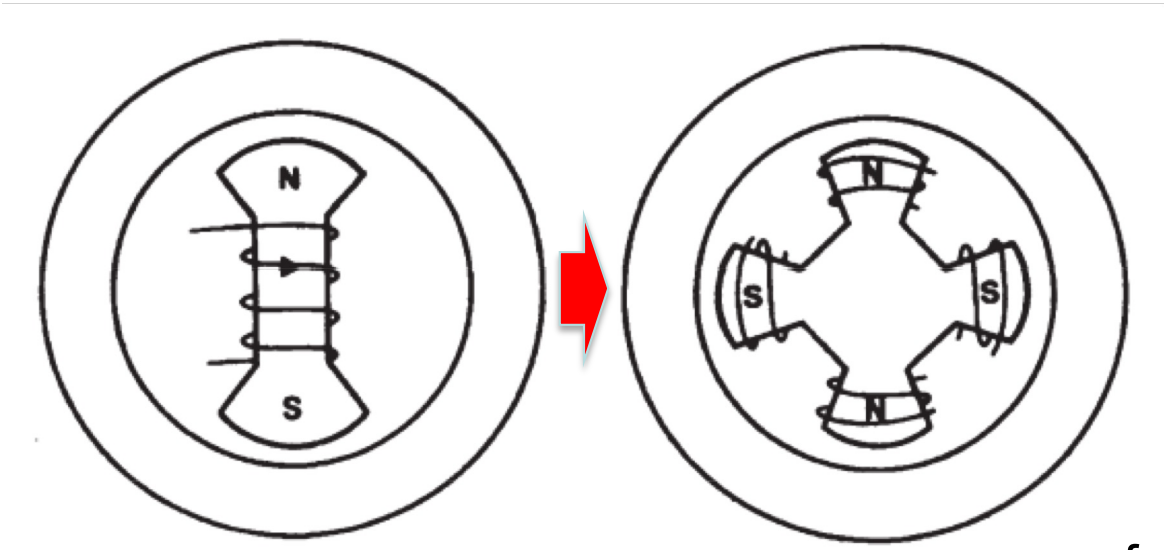
- Single dipole attached to the *rotor* will induce a voltage in *stator* coils.
- Arrangement of stator coils to produce 3 phase AC, commonly used for power transmission.
- Frequency will be equal to the rotor rotation frequency.





Generating Electricity

- Typical rotation speeds for large wind turbines are only 12-20 rpm.
- Need to deliver much higher AC frequencies (for connection to grid).
- Introduce *gearing* and/or increase *number of magnetic dipoles* :



rotor frequency

$$f_s = f_r \times n$$

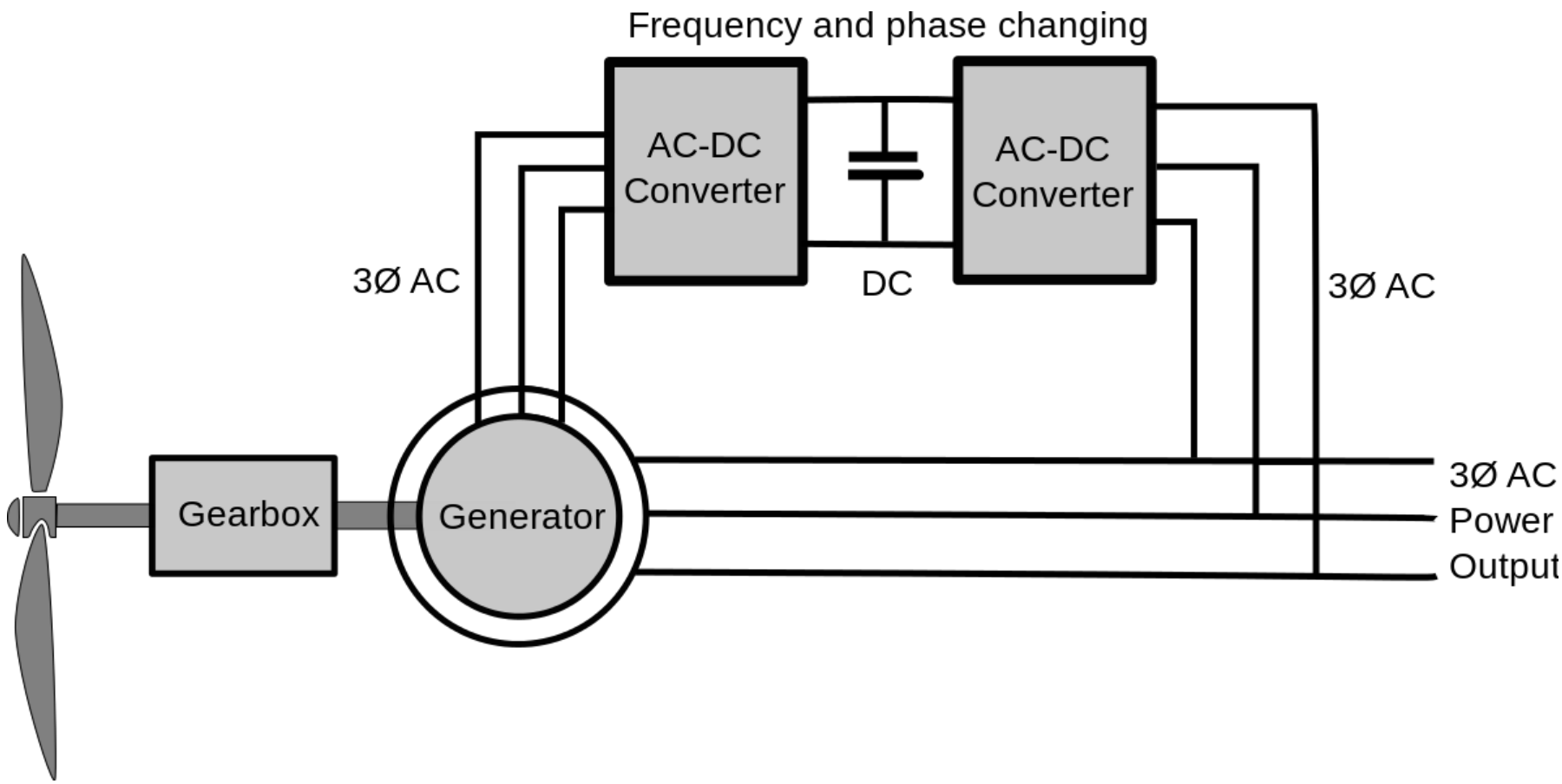
frequency of induced EMF in the stator coils (i.e. generated frequency)

number of pole pairs

FYI

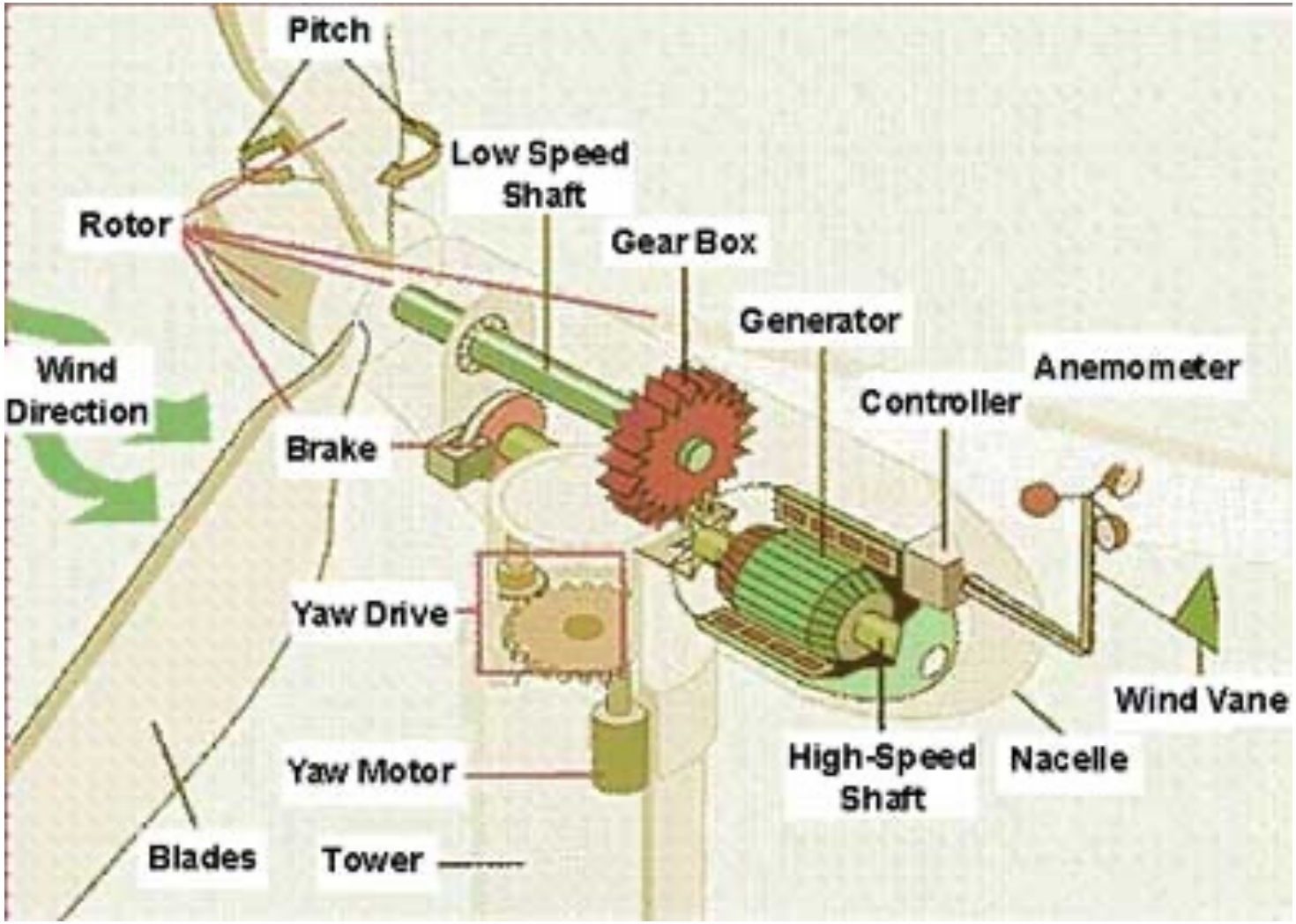
Generating Electricity

- Still need to accommodate variations in turbine rotation speeds.
- Most common method in modern large wind turbines is the use of *Doubly-Fed-Electrical Generators* :



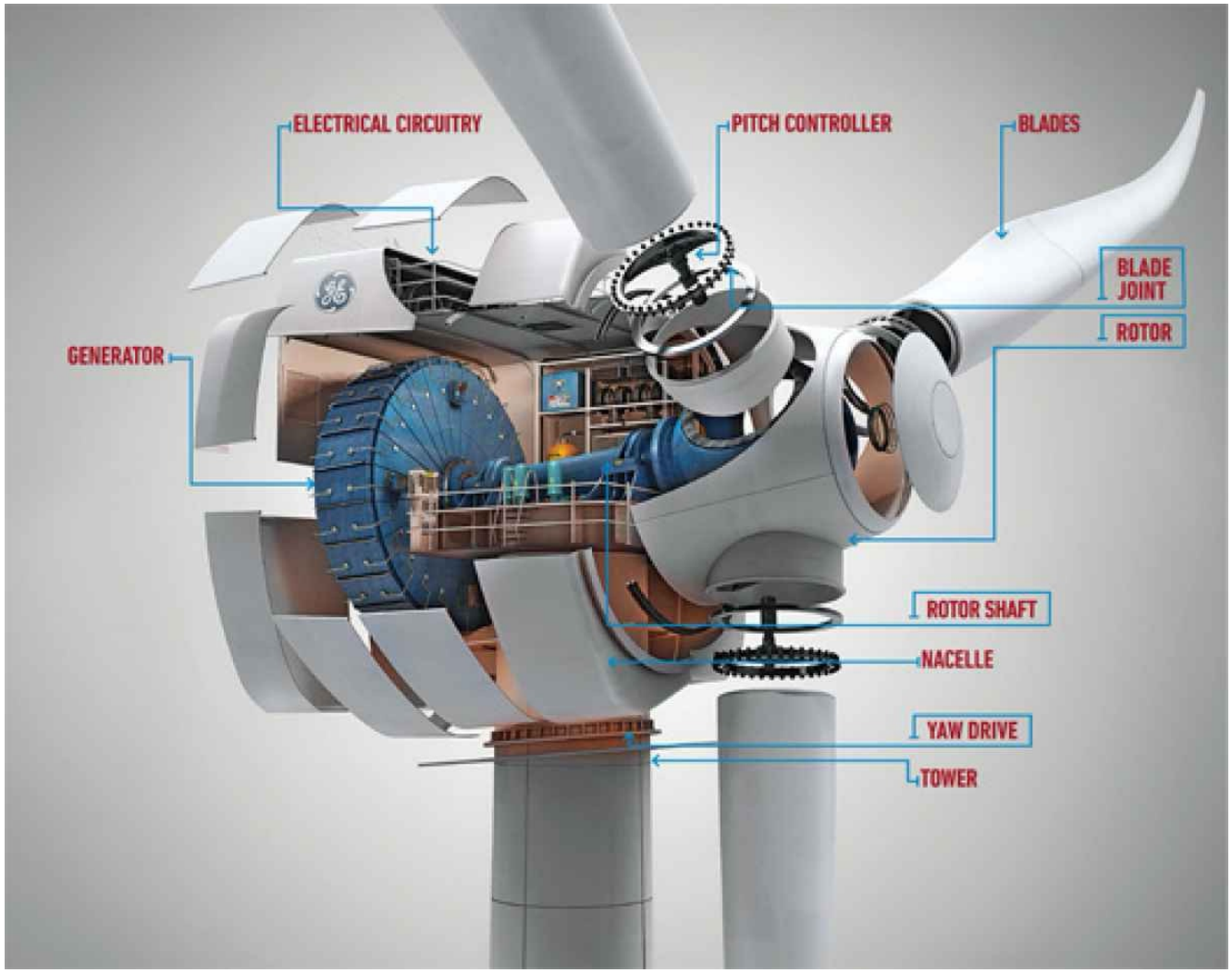
Turbine Generator with Gearbox

- Gear box to turn high-speed shaft at much larger frequency.
- Gear-boxes are heavy and always have an associated risk of mechanical failure.



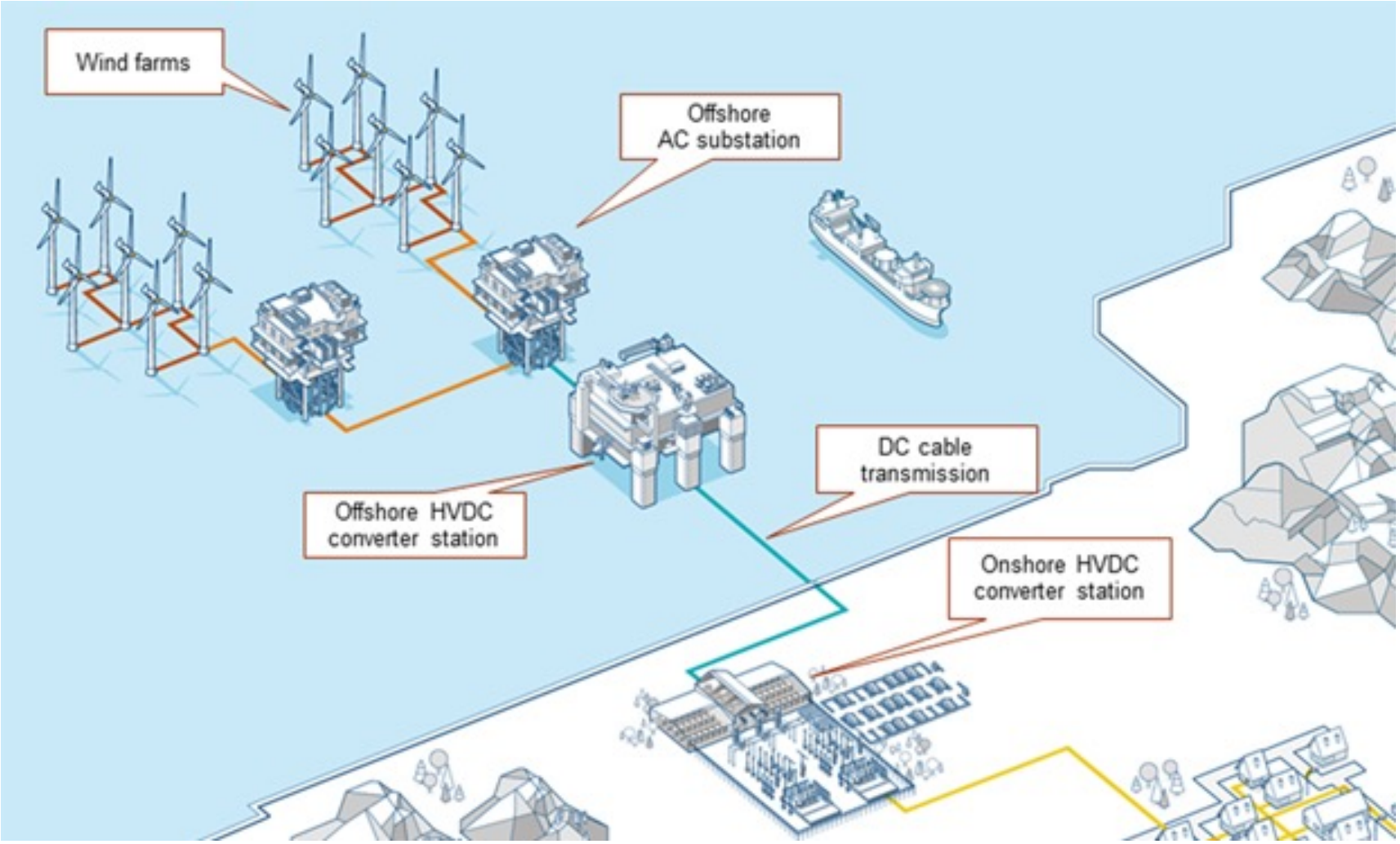
Direct Drive Generator

- Have a large number of magnetic poles on the rotor (e.g. permanent magnets)
- Generator must be large but no gear box.



Connecting to the Grid

- Onshore or short distance offshore : AC synchronised to grid.
- Long distance or far offshore : High-Voltage DC (HVDC; several 100 kV) to avoid capacitive losses. Requires rectification (AC➔DC) & conversion (DC➔AC).



Problem Based Learning Session 3

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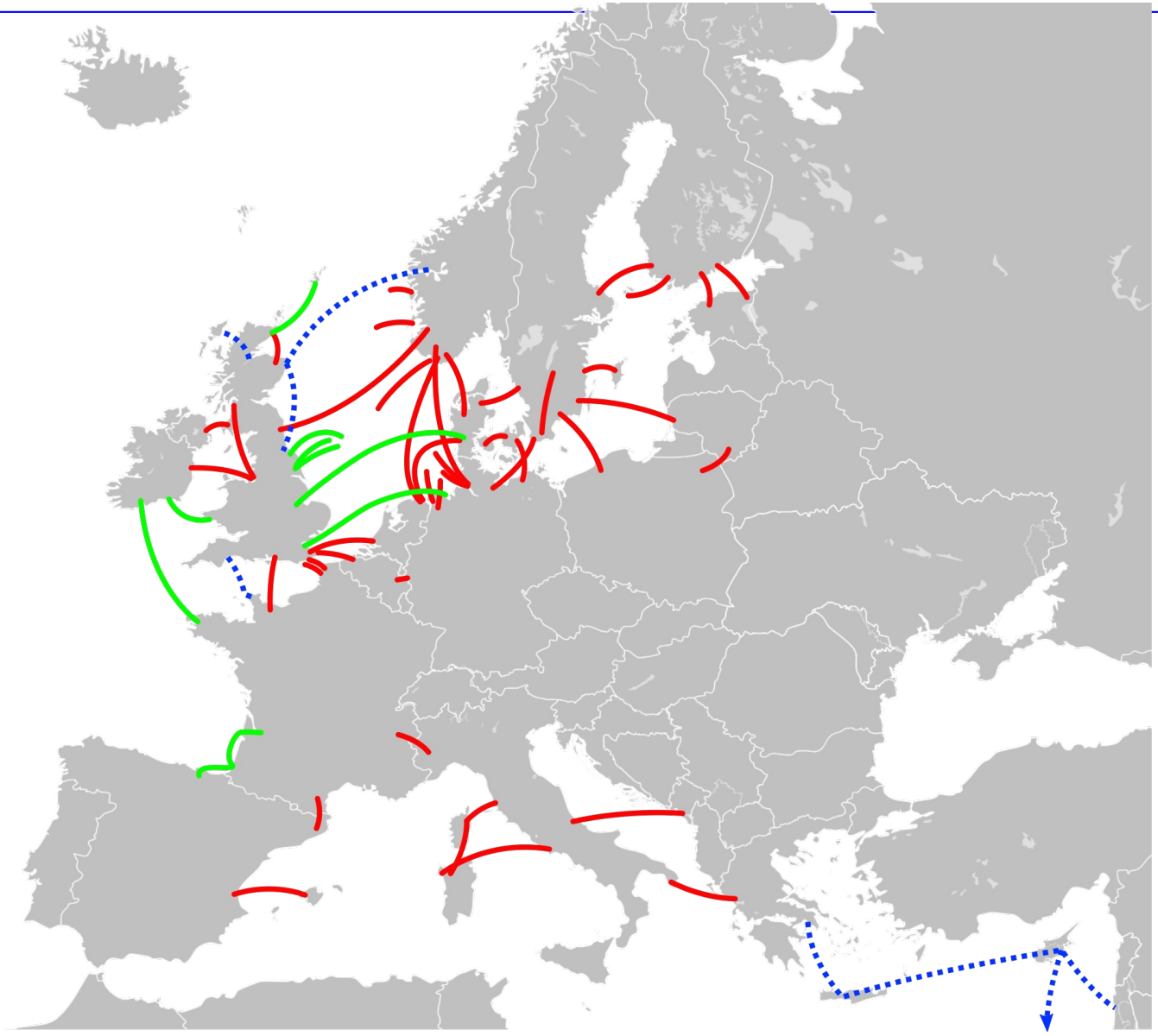
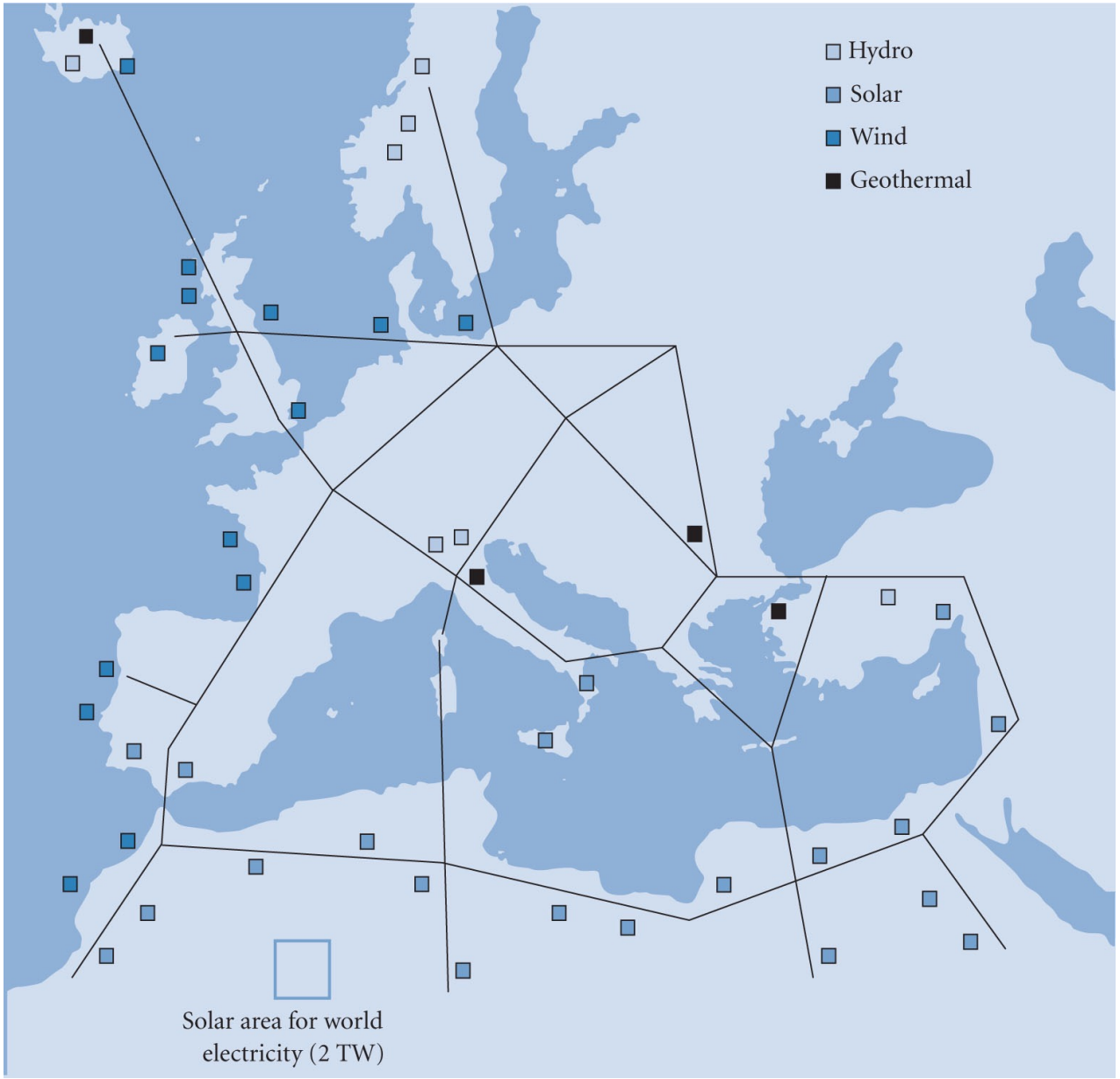
Interconnectors (and planned)



HVDC links in Europe Existing links Under construction Proposed Many of these HVDC lines transfer power from renewable sources such as hydro and wind. For names, see also the annotated version.[needs update]

https://en.wikipedia.org/wiki/Viking_Link

Saharan Solar



- "*Desertec*" proposal.
- HVDC grid linking renewable energy sources.
- Solar in desert regions makes a very large contribution to the total.

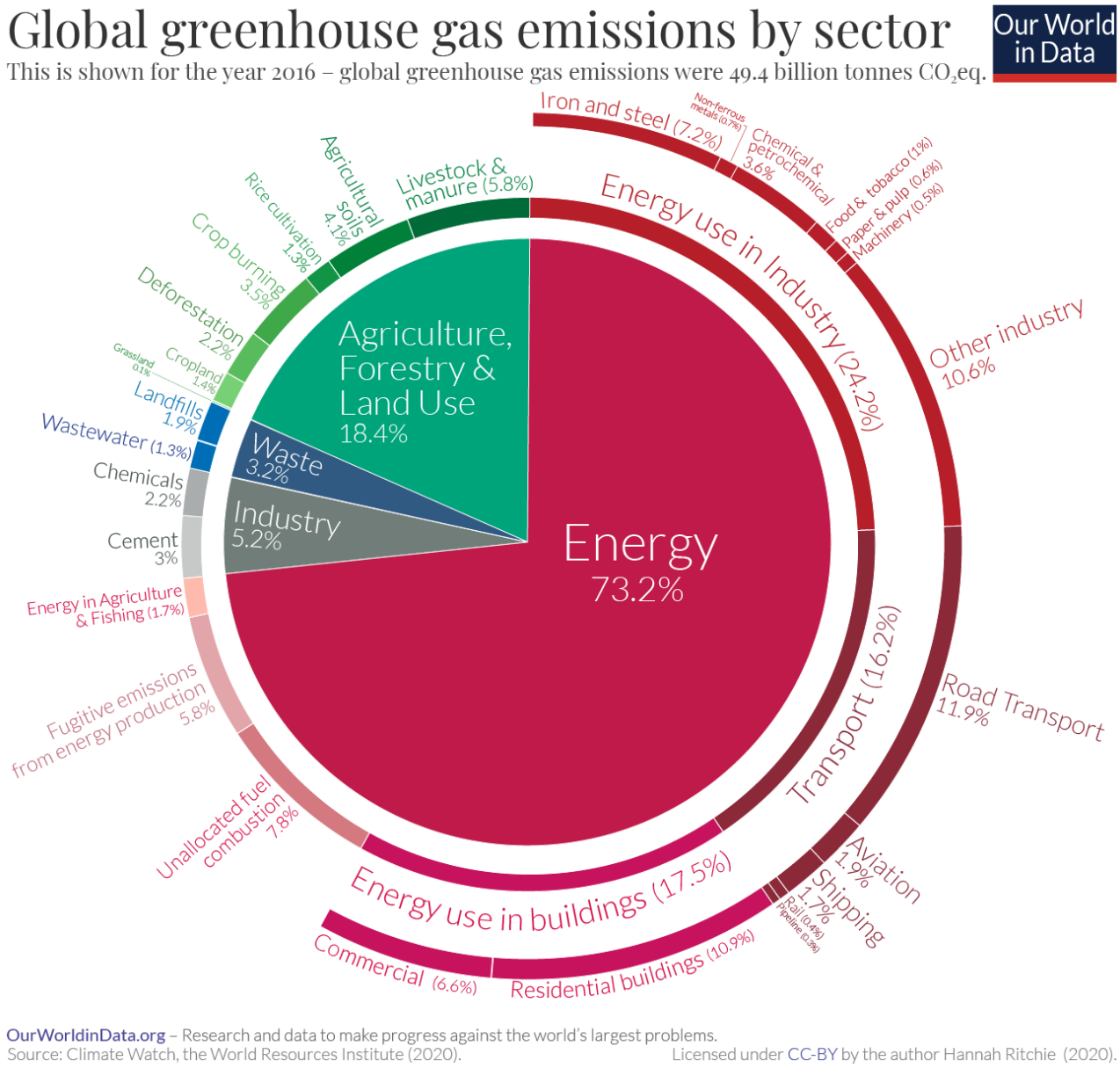
Xlinks

- https://xlinks.co/morocco-uk-power-project/
- https://en.wikipedia.org/wiki/Xlinks_Morocco–UK_Power_Project

2024 Q1d

There is currently a proposal to build a wind and solar plant in Morocco to supply the UK Grid with renewable electricity via an 3800 km HVDC interconnector that has a resistance of 0.01 Ohm/km. Calculate the fractional ohmic energy loss if this HVDC line is used to transmit 1.5 GW of electrical power at a voltage of 500 kV and has a resistance of 0.01 Ohm/km.

Identify a challenge to the feasiblity of such a project and suggest how it could be mitigated.


Global greenhouse gas emissions by sector
This is shown for the year 2016 – global greenhouse gas emissions were 49.4 billion tonnes CO_2eq.
Our World in Data
Energy 73.2%
Agriculture, Forestry & Land Use 18.4%
Industry 5.2%
Waste 3.2%
Energy use in Industry (24.2%)
Iron and steel (7.2%)
Non-ferrous metals (0.7%)
Chemical & petrochemical 3.6%
Food & tobacco (1%)
Paper & pulp (0.6%)
Machinery (0.5%)
Other industry 10.6%
Transport (16.2%)
Road Transport 11.9%
Aviation 1.9%
Shipping 1.7%
Rail (0.4%)
Pipeline (0.3%)
Energy use in buildings (17.5%)
Residential buildings (10.9%)
Commercial (6.6%)
Unallocated fuel combustion 7.8%
Fugitive emissions from energy production 5.8%
Energy in Agriculture & Fishing (1.7%)
Cement 3%
Chemicals 2.2%
Wastewater (1.3%)
Landfills 1.9%
Grassland 0.1%
Cropland 1.4%
Deforestation 2.2%
Crop burning 3.5%
Rice cultivation 1.3%
Agricultural soils 4.1%
Livestock & manure (5.8%)
OurWorldinData.org – Research and data to make progress against the world's largest problems.
Source: Climate Watch, the World Resources Institute (2020).
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